



Slice combination convolutional neural network based radio frequency fingerprint identification for Internet of Things

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Abstract

Deep learning approaches offer new ideas for individual identification of communication radiation sources. It is an important issue for researchers to establish a suitable deep learning model for radio frequency fingerprint identification under deep learning based communication framework. To effectively identify IoT objects, and prevent the occurrence of problems such as device cloning, replay attacks and user identity impersonation, given the limitations of existing methods, a novel slice combination convolutional neural network model is designed for individual identification of communication radiation sources. First, radio frequency baseband I/Q signal is collected by a receiver; And then steady-state segment of the I/Q signal is intercepted through variance trajectory detection and divided into many slices; At last, the fingerprint of each radio frequency baseband I/Q signal, which involves physical layer essential properties of the transmitter, can be effectively extracted and learned by the proposed slice combination convolutional neural network model, and the individual recognition of the transmitter can be finally realized. Experimental test on the ADS-B radio frequency fingerprint recognition from 198 civil aviation airplanes in the real-word illustrate that the proposed approach can greatly improve the recognition accuracy compared with existing machine learning based and deep learning based methods, which is very suitable for large-scale real-world individual identification of communication radiation sources.

Keywords Internet of Things · Radio frequency fingerprint · Deep learning · Large-scale · ADS-B signal

Abbreviations

RF	Radio frequency
RFF	Radio frequency fingerprint
IoT	Internet of Things
SEI	Specific emitter identification
PSD	Power spectral density
GRA	Gray relation algorithm
SVM	Support vector machine
SNR	Signal-to-noise ratio
DAC	Digital-to-analog converters
ICAO	International Civil Aviation Organization

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1 Introduction

Information security is critical for establishing a robust IoT. Information security issues dominated by wireless communication networks are emerging due to openness of radio transmission, for example, device cloning, replay attacks and user identity impersonation [1–3]. Reliable identification and certification is essential to ensure information security of IoT devices. IoT objects should have their own identities to build a credible IoT ecological network system [4, 5]. In order to prevent the occurrence of problems such as device cloning, replay attacks and user identity impersonation, how to accurately identify and authenticate IoT objects is a critical issue faced by the IoT, and is also the foundation of IoT applications [6–8]. Traditional authentication mechanisms use cryptographic algorithms to produce numerical results, that are difficult for third parties to forge, at the application layer. However, above-mentioned application layer authentication

mechanisms usually have problems such as protocol security loopholes and key leakage. Nowadays, IoT devices often have properties of diversification, intelligence and complexity. It has been hard for traditional authentication mechanism to satisfy IoT information security requirements [9]. RFF identification is a non-cryptographic authentication mechanism based on the hardware properties of IoT objects, with no need of embed additional hardware or additional computing resources [10]. Individual identification of communication radiation sources on the basis of subtle characteristics of RF signal originated from SEI, which is the ability to associate the unique electromagnetic characteristics of the transmitter with the transmitter itself [11].

As illustrated in Fig. 1, in the light of the usage of physical layer resources, existing RFF recognition methods can be classified into channel characteristics based methods [12, 13] and transmission signal based methods. The channel characteristics based method aims to take IoT device unique location information as the identity detection indicator, and is always used to the IoT device indoor positioning. The transmission signal based method can be classified into transient signal based methods [14, 15] and steady-state signal based methods [16, 17]. RF transient signals do not involve data information, and only mirror transmitter hardware properties, and therefore RFF is originally extracted from RF transient signal. However, RF transient signals are difficult to capture due to their short duration. RF steady state signals are easier to obtain with a cheap receiver due to its long duration, but the RFF is more troublesome to extract. With progress of RFF recognition techniques, research focus of scholars has gradually shifted from using RF transient signals to using the leading sequence of RF signals, and then shifted to using transmission data fragment of RF steady-state signals, gradually reducing the requirements for

detection and interception of the required signal fragments to be identified. In addition, according to different fingerprint feature extraction methods, the transmission signal based RFF recognition method can also be classified into waveform domain based method [18] and modulation domain based method [19]. As in communication signal, quadrature modulation has been widely used nowadays, the modulation domain method adopts signal structure granted by the modulation scheme. And RF baseband I/Q signals are used as the basic processing unit to make transmitter specific physical properties easier to recognize. Modulation domain fingerprint features include constellation trace figure [20], differential constellation trace figure [21], contour stellar [19], differential contour stellar [22] and other features. During identification & authentication stage, the transmission signal based RFF recognition method can also be classified into traditional machine learning based method [23] and deep learning based method [24] according to different classifiers. Deep learning approaches offer new ideas for individual identification of communication radiation sources [25]. It is an important issue for researchers to establish a suitable deep learning model for radio frequency fingerprint identification under deep learning based communication framework. To prevent the occurrence of problems such as device cloning, replay attacks and user identity impersonation, and effectively identify IoT objects, given the limitations of existing methods, a novel slice combination convolutional neural network model is designed for large-scale individual identification of communication radiation sources. The main contents of this article is illustrated as follows:

- (1) Properties and types of radio frequency fingerprint identification technology are described.
- (2) The shortcomings of existing fingerprint recognition technology based on traditional machine learning are described with involvement of detailed experimental tests.
- (3) The shortcomings of existing fingerprint identification technology based on deep learning are described with involvement of detailed experimental tests.
- (4) Given the limitations of existing methods, a novel slice combination convolutional neural network model is designed for large-scale individual identification of communication radiation sources.

The rest of this article is arranged as follows. The shortcomings of existing traditional machine learning based fingerprint recognition technology are described in Section II, and the shortcomings of existing deep learning based fingerprint identification technology are also described. In Section III, the description of the proposed method is made. And experimental tests on the ADS-B radio frequency fingerprint recognition from 198 civil aviation

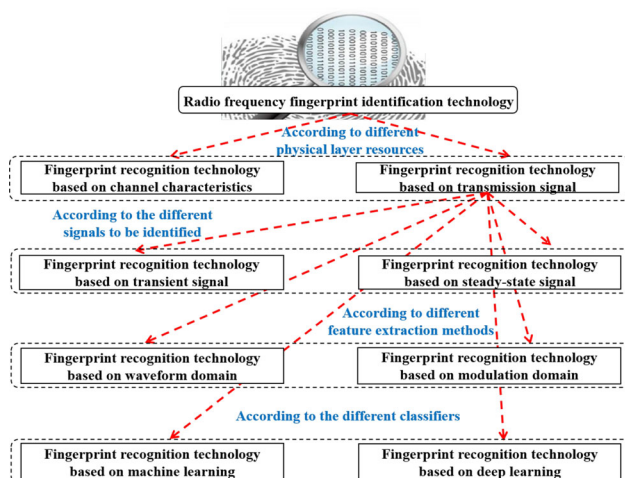


Fig. 1 Properties and types of radio frequency fingerprint identification technology

airplanes in the real-word are illustrated in Section IV, followed by the conclusion and discussion in Section V.

2 Problem description of existing methods

2.1 Machine learning based radio frequency fingerprint recognition method

Judging from the current research status of RFF recognition, it is still a challenging task to extract radio frequency fingerprint with unique native attributes. For fingerprint recognition technology based on traditional machine learning, feature extraction methods play a vital role in recognition accuracy. Common radio frequency fingerprint features include entropy feature [26], Holder coefficient feature [16], multifractal dimension feature [7], fractal box dimension feature [9], RF-DNA [27], integral bispectral feature, power spectral density feature, etc. Among them, Integral bispectrum and power spectral density are the two relatively best methods for extracting radio frequency fingerprint features in steady-state signals. In order to pursue higher data rates and spectrum efficiency, communication systems generally use linear modulation modes, such as QPSK and 16QAM. In a multi-carrier system, the peak-to-average ratio is large and the signal envelope changes, so the system should maintain linear amplification. Nonlinear amplification will cause in-band signal distortion, degrading system performance, resulting in out-of-band intermodulation products, and interference in adjacent channels of transmitter carrier frequency. The amplifier nonlinearity is mainly reflected in signal power spectrum, therefore the power spectral density feature in the energy domain can be extracted by power spectral estimation method.

The specific RFF recognition process based on the power spectral density fingerprint feature is shown in Fig. 2 as follows.

- I. First, the RFF baseband I/Q signal is collected by receiver, and the In-phase signal is selected;

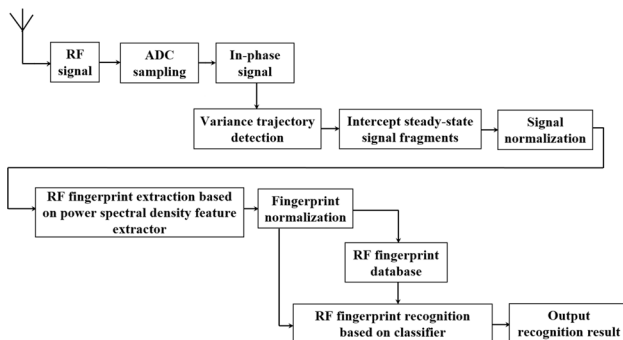


Fig. 2 RFF recognition based on power spectral density fingerprint feature and intelligent classifier

- II. Then the steady-state In-phase signal segment is intercepted through variance trajectory detection, and data normalization processing on the steady-state signal segment is performed;
- III. The power spectral density of the steady-state in-phase signal segment is calculated after data normalization processing to obtain feature vectors, and these feature vectors are regarded as the transmitter RFF;
- IV. Generally, the Yule-Walker power spectrum estimation method is used for power spectral density feature extraction. This method, also known as AR model parameter estimation method, is a classic modern power spectrum estimation method.
- V. Finally, an intelligent classifier is applied to recognize the RFF to complete individual identification of the communication radiation source after fingerprint normalization.

Intelligent classifiers include support vector machine and gray relation classifier. Take the GRA for classification and recognition as an example:

Define the power spectral density feature vector, extracted from RF signals to be recognized, to characterize the IoT device physical layer characteristics, as described as follows:

$$B_1 = \begin{bmatrix} b_1(1) \\ b_1(2) \\ b_1(3) \\ \dots \\ b_1(K) \end{bmatrix}, B_2 = \begin{bmatrix} b_2(1) \\ b_2(2) \\ b_2(3) \\ \dots \\ b_2(K) \end{bmatrix}, \dots, B_i = \begin{bmatrix} b_i(1) \\ b_i(2) \\ b_i(3) \\ \dots \\ b_i(K) \end{bmatrix}, \dots \quad (1)$$

where $B_i(i = 1, 2, \dots)$ is power spectral density feature vector, which represents the IoT device physical layer characteristics to be identified.

Define the established RFF library between the IoT device physical layer characteristics and the IoT device tags, as described as follows:

$$C_1 = \begin{bmatrix} c_1(1) \\ c_1(2) \\ c_1(3) \\ \dots \\ c_1(K) \end{bmatrix}, C_2 = \begin{bmatrix} c_2(1) \\ c_2(2) \\ c_2(3) \\ \dots \\ c_2(K) \end{bmatrix}, \dots \quad (2)$$

$$C_j = \begin{bmatrix} c_j(1) \\ c_j(2) \\ c_j(3) \\ \dots \\ c_j(K) \end{bmatrix}, \dots, C_m = \begin{bmatrix} c_m(1) \\ c_m(2) \\ c_m(3) \\ \dots \\ c_m(K) \end{bmatrix}$$

where $C_j (j = 1, 2, \dots)$ is the IoT device tag of known and legal IoT device, and $c_j (j = 1, 2, \dots)$ is a certain RFF characteristic parameter.

Define $\rho \in (0, 1)$:

$$\xi(b_i(k), c_j(k)) = \frac{\min_j \min_k |b_i(k) - c_j(k)| + \rho \cdot \max_j \max_k |b_i(k) - c_j(k)|}{|b_i(k) - c_j(k)| + \rho \cdot \max_j \max_k |b_i(k) - c_j(k)|} \quad (3)$$

$$\xi(B_i, C_j) = \frac{1}{K} \sum_{k=1}^K \xi(b_i(k), c_j(k)), \quad j = 1, 2, \dots \quad (4)$$

where ρ is resolution coefficient, and the value is always 0.5; $\xi(b_i(k), c_j(k))$ represents correlation coefficient of k_{th} RFF feature parameter between B_i and C_j , and $\xi(B_i, C_j)$ represents gray relation degree between B_i and C_j .

When $\xi(B_i, C_j)$ ($j = 1, 2, \dots, m$) between B_i and each C_j ($j = 1, 2, \dots, m$) in RFF library is obtained, in order to identify whether the communication radiation source individual to which B_i belongs is a legal wireless communication device, the following judgment criteria is added here:

$$test = \left(\frac{\max(\xi(B_i, C_j), j = 1, 2, \dots, m) - \frac{1}{m}}{\sum_{j=1}^m \xi(B_i, C_j)} - \frac{1}{m} \right) / \left(\frac{1}{m} \right) \quad (5)$$

If $test$ is less than a certain threshold (the defined threshold is 0.0042), the communication radiation source individual to which B_i belongs is an illegal wireless communication device; Otherwise, the communication radiation source individual to which B_i belongs is a legal wireless communication device, and B_i can be classified into the communication radiation source individual tag to which the highest gray relation degree belongs in the radio frequency fingerprint library.

The individual characteristic information of the radiation source is attached to the emission signal, and it is difficult to directly extract the individual characteristics. The machine learning based radio frequency fingerprint recognition method usually transforms signals into different transform domains, and extracts multi-domain subtle features in different transform domains, to increase the amount of information about technical differences between signal samples, and enhance individual characteristics. Due to that there are many types and quantities of terminals, and new products are constantly entering the IoT network, and the traditional RF fingerprint characteristics are unstable, and therefore, another way is needed to achieve the goal as there are many factors that affect the recognition performance.

2.2 Deep learning based radio frequency fingerprint recognition method

As illustrated in Fig. 3, the transmitter imperfections, which exist in electronic components, is the physical basis of RFF identification.

As in communication signal, quadrature modulation has been widely used nowadays, the modulation domain method adopts signal structure granted by the modulation scheme. And RF baseband I/Q signals are used as the basic processing unit to make transmitter specific physical properties easier to recognize, as seen in Fig. 4.

Deep learning approaches offer new ideas for RFF identification. However, the current fingerprint recognition technology based on deep learning mainly uses the RF baseband data directly as training data, trying to let the algorithm find the fingerprint features by itself, and has achieved a certain effect. However, due to the its characteristics of black box, it is best to combine it with the method of feature engineering to enhance the interpretability of the deep learning model and improve the understanding of the mechanism of RF fingerprinting, as seen in Fig. 5.

At this stage, the work of intelligently and automatically extracting fingerprint features is still in the stage of trial and exploration. In particular, how to make better use of the advantages of intelligent computing combined with expert experience, and how to extract the truly stable and effective fingerprint characteristics of radiation sources requires more detailed and in-depth research. The robustness of RF fingerprints is a key requirement for RF fingerprints to become practical, and there is still a lot of detailed work to be done.

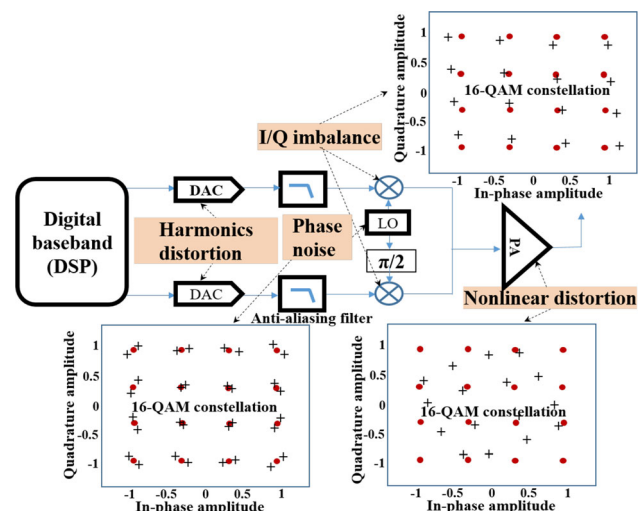


Fig. 3 Transmitter imperfections [28]

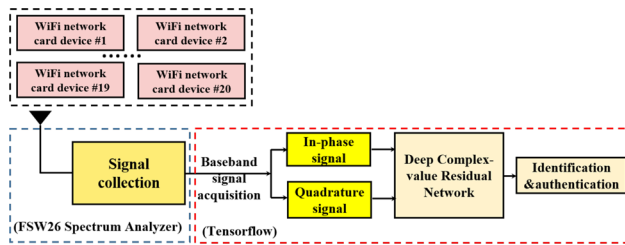


Fig. 4 Radio frequency fingerprint identification based on deep complex residual network [19]. *Note* The principle of radio frequency fingerprint identification based on deep complex residual network in Fig. 4 can be referred to our previous work [19]

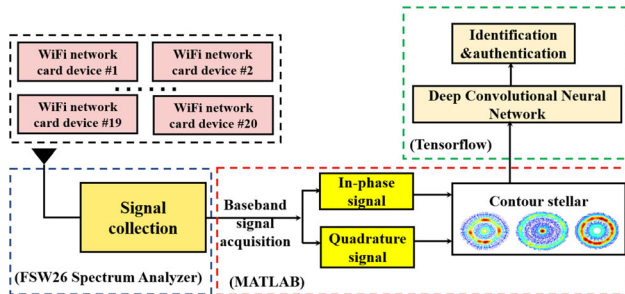


Fig. 5 Contour stellar based radio frequency fingerprint identification [22]. *Note* The principle of contour stellar based radio frequency fingerprint identification in Fig. 5 can be referred to our previous work [22]

3 The proposed method

The individual identification of the radiation source based on the RF signal fingerprint is based on the difference of the physical layer hardware of the device, which theoretically does not change with the change of the transmitted signal. Just like each person has a different fingerprint, each wireless device also has a different RF fingerprint, and this difference is reflected in the communication signal, which can be extracted by analyzing the received RF signal. However, with the development of the hardware product industry, the radio frequency device technology of terminal equipment is continuously improved, and the tolerance is gradually narrowed, which makes the extraction and distinction of radio frequency fingerprints more difficult. The future RF fingerprint features need to develop in the direction of more refinement, intelligence, complexity and deeper level. In addition, many deep learning models applied in the field of communication are designed based on general models, such as convolutional neural networks are often used in image classification problems, while recurrent neural networks are often used in the field of natural language processing. Although the deep learning models commonly used in the field of computer science can be applied to the field of communication. However, in actual communication engineering projects, establishing a

general deep learning model suitable for communication scenarios is not only conducive to optimizing the communication system, but also reducing the cost and time of model selection. Therefore, it is an important issue for researchers to establish a suitable deep learning model for radio frequency fingerprint identification under deep learning based communication framework.

In order to guide the deep neural network structure to learn the real RF fingerprint characteristics, from two aspects of signal slice combination, feature combination and fusion, a novel slice combination convolutional neural network model is designed for large-scale individual identification of communication radiation sources, as illustrated in Fig. 6.

To prevent the occurrence of problems such as device cloning, replay attacks and user identity impersonation, and effectively identify IoT objects, the method proposed in this paper extracts the subtle features on different slices through deep convolutional neural networks, and then increases the amount of information on the technical differences between RF signal samples through feature combination and fusion, thereby enhancing individual features.

First, radio frequency baseband I/Q signal is collected by a receiver. And then steady-state segment of the I/Q signal is intercepted through variance trajectory detection and divided into many slices, as show in Fig. 7.

For signal sequence $\{x(k)\}$, $k = 1, 2, \dots, N_x$, the variance trajectory sequence $\{VT_x(i)\}$ of the signal sequence can be calculated by the following two equations:

$$VT_x(i) = |W_x(i) - W_x(i-1)|, \quad i = 2, 3, \dots, L \quad (6)$$

$$W_x(m) = \frac{1}{N_w} \sum_{k=1+(m-1)N_s}^{1+(m+1)N_s+N_w} [x(k) - \mu_w]^2, \quad m = 1, 2, \dots, L \quad (7)$$

where $L = \lfloor (N_x - N_w)/N_s \rfloor + 1$, and $\lfloor \cdot \rfloor$ means round down; N_w represents the length of the sliding window used for variance calculation; N_s represents the displacement of the window function after each calculation; μ_w represents the mean of the window function sequence. When a transient signal is acquired by the receiver, the magnitude of

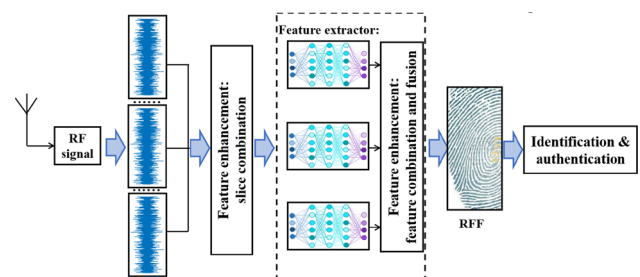


Fig. 6 The proposed method for large-scale individual identification of communication radiation sources

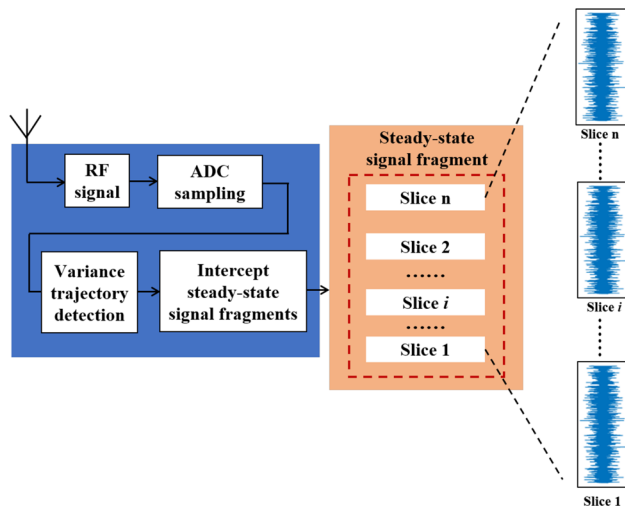


Fig. 7 Signal acquisition and slicing

$W_x(i)$ will increase significantly. After the variance trajectory calculation is completed, the problem of starting point detection of transient signal can be simplified to the common change point detection problem. The change point position where the variance trajectory sequence begins to rise is the starting point of the transient signal. Similarly, when the variance trajectory begins to remain unchanged, it is the position where the transient signal ends, and the signal power is basically stable at this time. The variance trajectory detection method has low computational complexity, and is suitable for real-time detection of signal change points and interception of identifiable signals.

At last, the fingerprint of each radio frequency baseband I/Q signal, which involves physical layer essential properties of the transmitter, can be effectively extracted and learned by the proposed slice combination convolutional neural network model, seen in Fig. 8, and the individual recognition of the transmitter can be finally realized.

As shown in Fig. 8, in the slice combination convolutional neural network model, the structure of deep convolutional neural network feature extractor is shown in Table 1.

The activation function of the convolutional layer (Conv2D) is ReLU, as follows:

$$f(x) = \max(0, x) \quad (8)$$

The calculation process of the Batch normalization layer (BN) is as follows:

$$\hat{x}^{(k)} = \frac{x^{(k)} - E[x^{(k)}]}{\sqrt{\text{Var}[x^{(k)}]}}, \text{BN}(\hat{x}^{(k)}) = \gamma^{(k)}\hat{x}^{(k)} + \beta^{(k)} \quad (9)$$

Using Batch normalization layer (BN) can effectively reduce the dependence of the deep network on weight initialization.

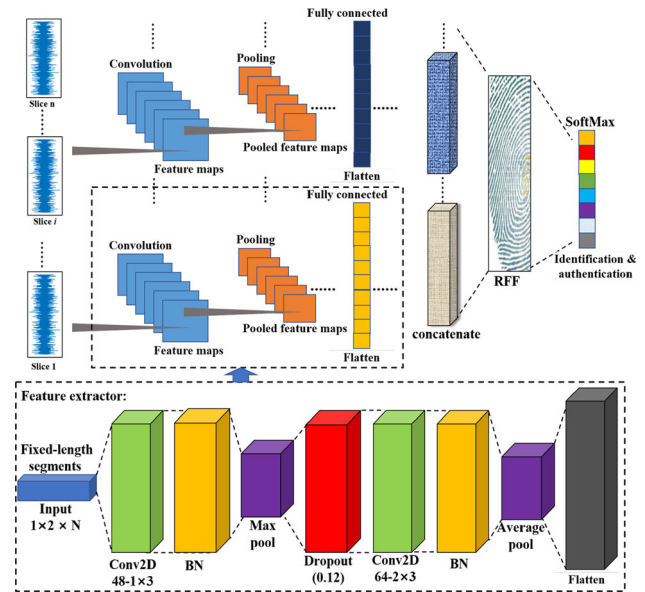


Fig. 8 The proposed slice combination convolutional neural network model

Table 1 The structure of deep convolutional neural network feature extractor

Layer	Output shape
Input	$1 \times 2 \times N$
Convolution2D	$1 \times 2 \times 48$
BatchNormalization	$1 \times 2 \times 48$
MaxPooling2D	$1 \times 1 \times 48$
Dropout	$1 \times 1 \times 48$
Convolution2D	$3 \times 3 \times 64$
BatchNormalization	$3 \times 3 \times 64$
AveragePooling2D	$3 \times 1 \times 64$
Flatten	198

N is the fixed length of the signal segment

The activation function of the output layer is Softmax, as follows:

$$\sigma(z_j) = \frac{e^{z_j}}{\sum_{i=1} e^{z_i}} \quad (10)$$

The loss function of the deep network is sparse_categorical_crossentropy. A large amount of sparsity is introduced into the deep network, such as ReLU, Dropout, Pooling, etc., which may lead to a decrease in the computational performance of the deep network. In order to maintain the sparsity of the network structure and utilize the high computational performance of dense matrices, the design of slice combination convolutional neural network can cluster sparse matrices into denser sub-matrices to

improve computational performance. The network learning rate is 0.0001, and the Adam optimizer is used to update the network weight.

4 Experiment and analysis

4.1 Experiment introduction

We choose ADS-B signals as test signals for large-scale real-world individual identification of communication radiation sources for the following reasons: large scale, easy labeling and open source. The ADS-B system installed on the airplane uses GPS to determine its location, and then transmits data such as its location, identity, altitude, and speed to the outside through a transmitter at a rapid time interval. The dedicated ADS-B ground station can receive the transmitted data and forward it to the air traffic controller to accurately track the airplane. At present, the airplane's identity authentication is mainly through information authentication after demodulating the ADS-B signals. A SM200B spectrum analyzer is taken as receiver to collect the RF baseband I/Q signals, and test environment is in real-world. And the detailed experimental devices and data collection process can be seen in reference [29]. Table 2 shows the parameter settings of the receiver.

ADS-B signals from 198 civil aviation airplanes are used to create a dataset. We collect about 200–600 signal samples for each ADS-B device, i.e., this is a situation where the sample size is uneven. Each signal sample contains 3000 steady-state data points. Figure 9 shows 4 signal samples from 4 various airplanes.

The ratio of the number of training samples to test samples from the dataset is set to be 4:1. And each signal sample is divided into two slices. The fingerprint of each signal sample, which involves physical layer essential properties of the transmitter, can be effectively extracted and learned by the proposed slice combination convolutional neural network model, and the individual recognition of the transmitter can be finally realized.

Table 2 The parameter settings of the receiver

Sampling Frequency	50 M Samples/s
IF	0 Hz (baseband)
Center Frequency	1090 MHz
Bandwidth	10 MHz
Gain	30 dB

4.2 Test case 1

To test the effectiveness of the proposed method, we conduct three test case analyses. Test case 1 is to test the change effect of the recognition accuracy before and after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure.

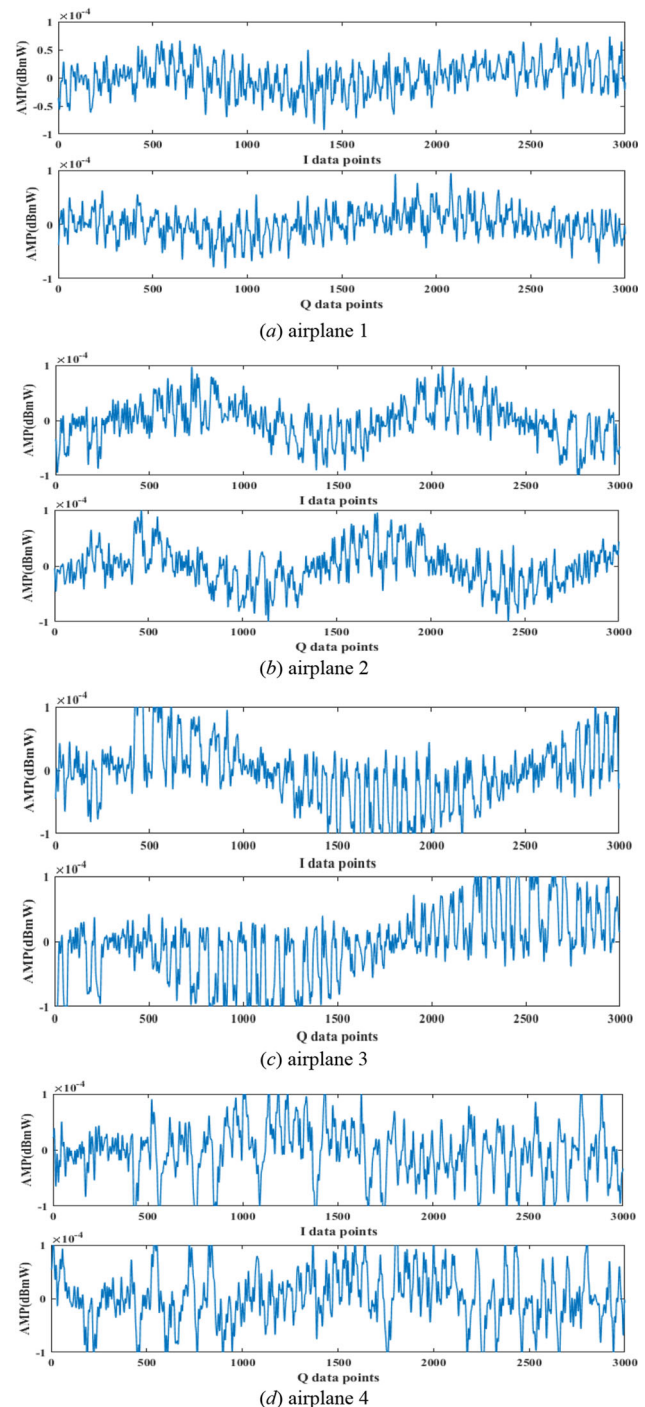


Fig. 9 4 signal samples from 4 various airplanes from the collected dataset

Test case 2 is to test the improvement of the recognition accuracy of the method proposed in this paper compared with the existing fingerprint recognition technology based on traditional machine learning. Test case 3 is to test the improvement of the recognition accuracy of the method proposed in this paper compared with the existing fingerprint recognition technology based on deep learning.

For test case 1, the recognition accuracy of 21 airplanes and 198 airplanes are obtained before and after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure, as shown in Figs. 10, 11, 12, and 13.

From Figs. 10 and 11, we can see that the recognition accuracy of 21 airplanes before slice combination and feature fusion is 92.68%, and the recognition accuracy of 21 airplanes after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure can be 99.70%.

From Figs. 12 and 13, we can see that the recognition accuracy of 198 airplanes before slice combination and feature fusion is only 74.60%, and the recognition accuracy of 198 airplanes after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure can be 99.03%.

The training convergence curve of 198 airplanes before and after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure is illustrated in Fig. 14.

From Fig. 14, for the deep network before slice combination and feature fusion, we can see that, about 350 training iterations, the training curve basically converges, the final training accuracy (acc-CNN) is 79.88%, and the prediction accuracy (val-acc-CNN) is 74.60%. For the deep network after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure, we can see that, about only 50 training

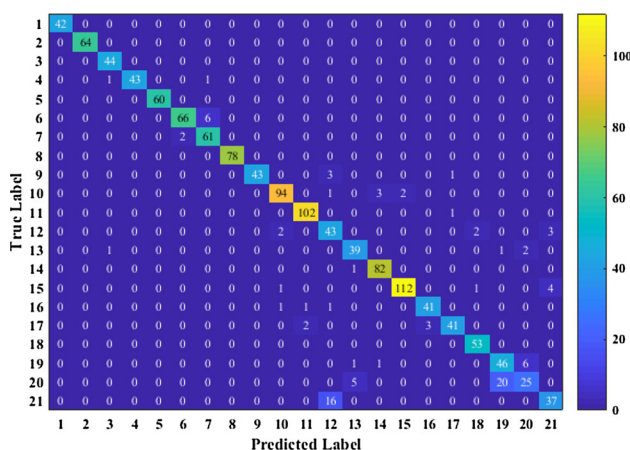


Fig. 10 the recognition accuracy of 21 airplanes before slice combination and feature fusion

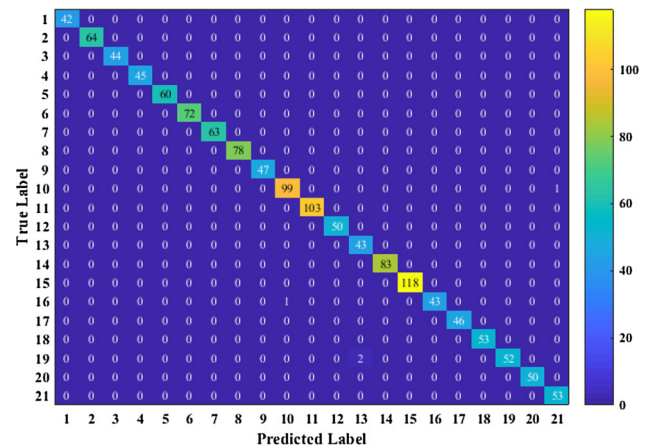


Fig. 11 the recognition accuracy of 21 airplanes after slice combination and feature fusion

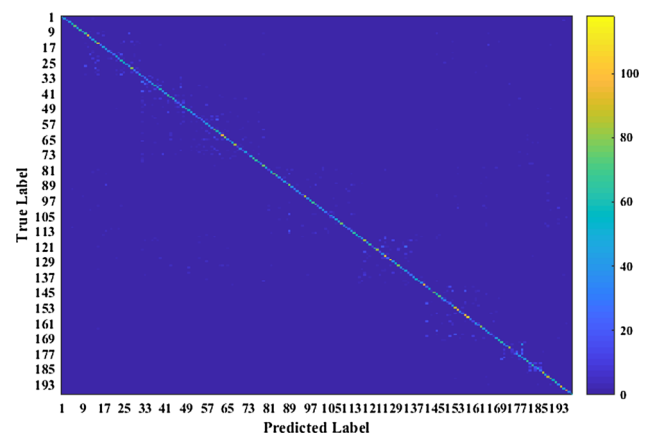


Fig. 12 the recognition accuracy of 198 airplanes before slice combination and feature fusion

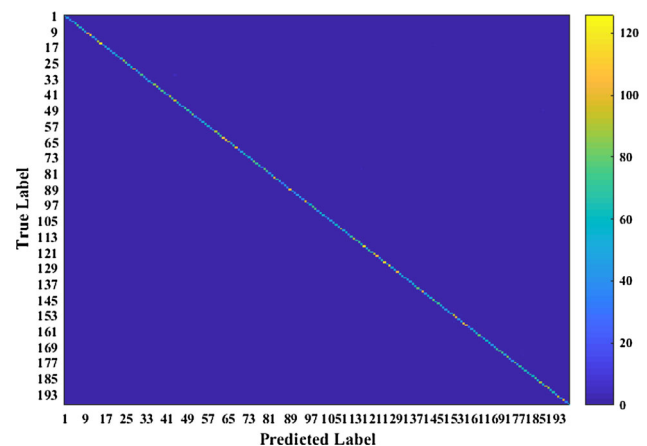


Fig. 13 the recognition accuracy of 198 airplanes after slice combination and feature fusion

iterations, the training curve basically converges, the final training accuracy (acc-SCCNN) is 99.55%, and the prediction accuracy (val-acc-SCCNN) is 99.03%. The test

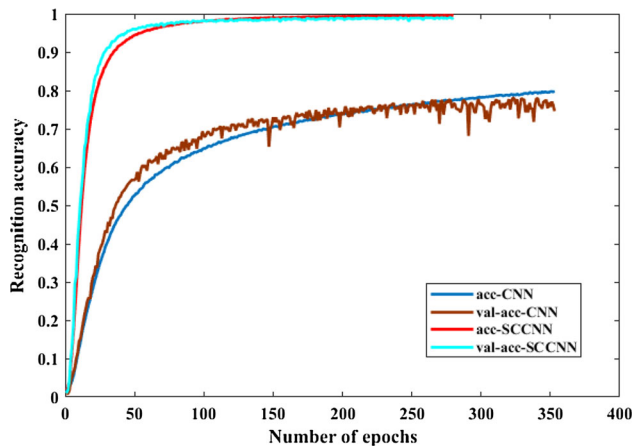


Fig. 14 The training convergence curve before and after slice combination and feature fusion under the same deep convolutional neural network feature extractor structure. *Note* CNN is the proposed method with only single slice, that is the deep network without slice combination and feature fusion; SCCNN is the proposed method with two slices, that is the deep network with combination of two slices and feature fusion under the same deep convolutional neural network feature extractor structure. Each signal slice contains 1500 steady-state data points

case 1 illustrated that the proposed method can effectively guide the deep neural network structure to learn the real RF fingerprint characteristics, by signal slice combination, feature combination and fusion.

4.3 Test case 2

Test case 2 is to test the improvement of the recognition accuracy of the method proposed in this paper compared with the existing fingerprint recognition technology based on traditional machine learning.

For test case 2, the recognition accuracy of 21 airplanes, 35 airplanes, 100 airplanes and 198 airplanes with PSD fingerprint and intelligent classifier are obtained as shown in Figs. 15, 16, 17, 18, 19, 20.

From Figs. 15 and 16, we can see that the recognition accuracy of 21 airplanes with PSD fingerprint and GRA classifier is 82.09%, and the recognition accuracy of 21 airplanes with PSD and SVM classifier can be 91.23%.

From Fig. 17, we can see that the recognition accuracy of 35 airplanes with PSD and SVM classifier is reduced to 83.35%.

From Fig. 18, we can see that the recognition accuracy of 100 airplanes with PSD and SVM classifier is reduced to 74.23%.

From Fig. 19, we can see that the recognition accuracy of 198 airplanes with PSD and SVM classifier is further reduced to 68.31%.

From Fig. 20, we can see that, when the number of airplanes increases, the recognition accuracy of this method

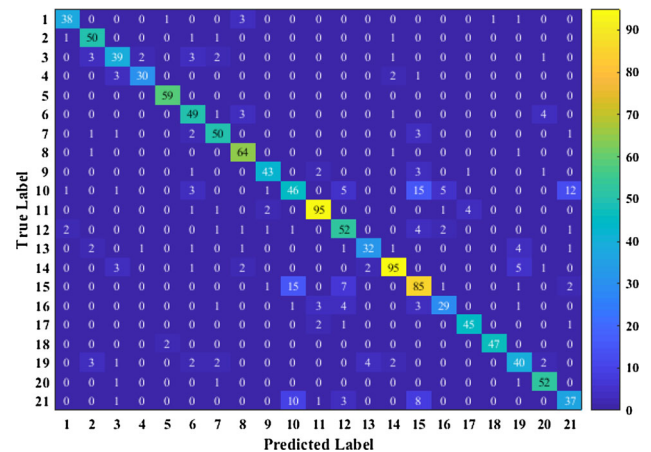


Fig. 15 The recognition accuracy of 21 airplanes on the basis of PSD fingerprint and GRA classifier

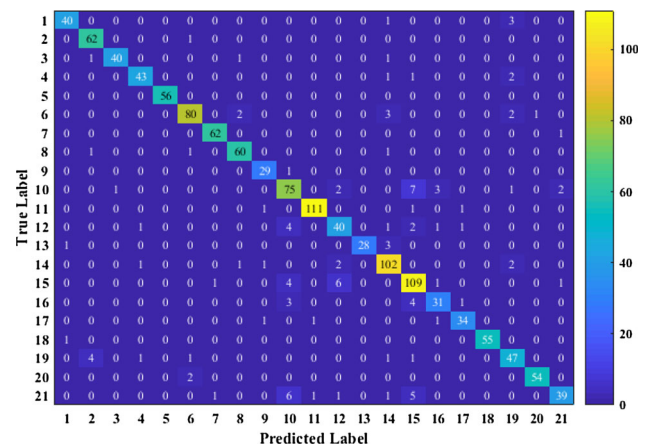


Fig. 16 The recognition accuracy of 21 airplanes on the basis of PSD fingerprint and SVM classifier

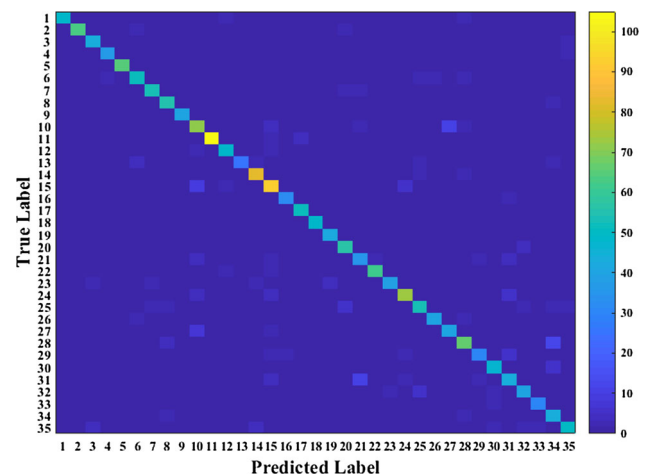


Fig. 17 The recognition accuracy of 35 airplanes on the basis of PSD fingerprint and SVM classifier

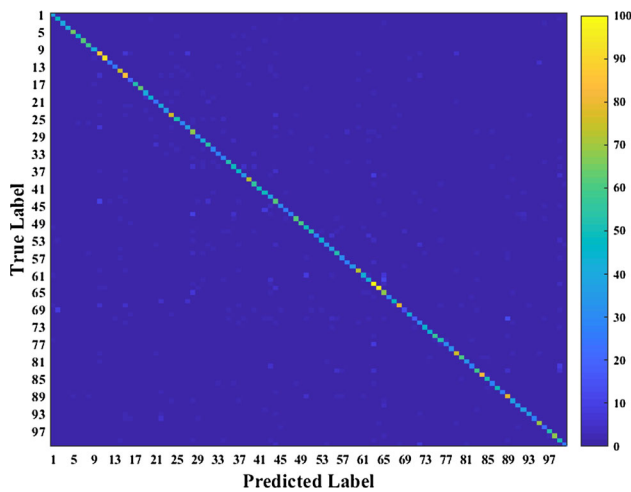


Fig. 18 The recognition accuracy of 100 airplanes on the basis of PSD fingerprint and SVM classifier

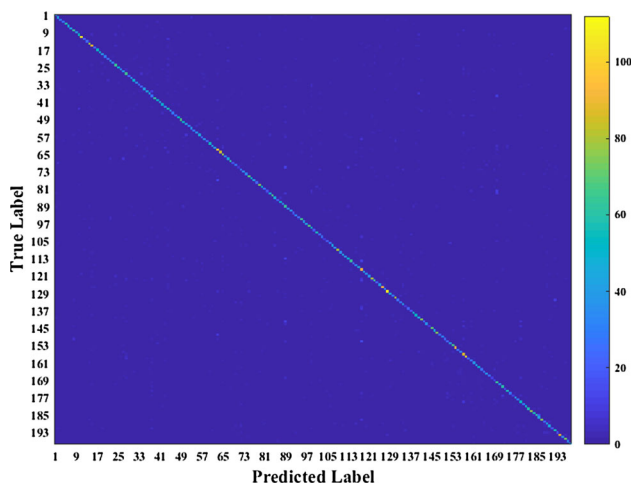


Fig. 19 The recognition accuracy of 198 airplanes on the basis of PSD fingerprint and SVM classifier

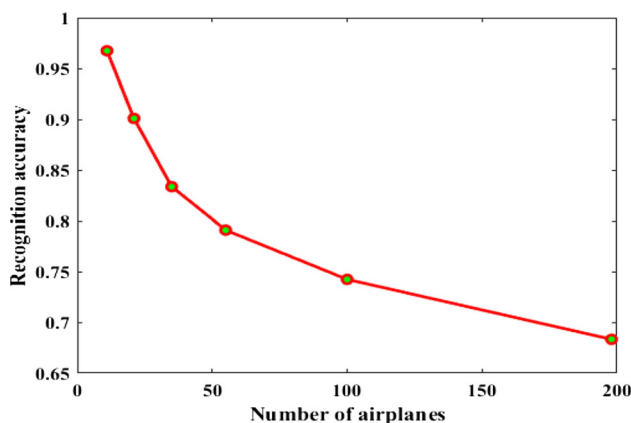


Fig. 20 The recognition accuracy with different number of airplanes

will decrease significantly, therefore the traditional machine learning based RFF recognition technology is not suitable for large-scale real-world radio frequency fingerprint identification for Internet of Things.

4.4 Test case 3

Test case 3 is to test the improvement of the recognition accuracy of the method proposed in this paper compared with the existing fingerprint recognition technology based on deep learning.

For test case 3, to test the improvement of the recognition accuracy of the method proposed compared with the existing fingerprint recognition technology based on deep learning, we choose contour stellar based method [22], deep complex residual network based method [19], complex convolutional neural network based method [30] as comparative algorithms. The confusion matrices of comparative recognition results by these methods are shown in Figs. 21, 22, 23, and 24, and the comparative recognition results by these methods are shown in Table 2.

From Fig. 21, we can see that the recognition accuracy of 21 airplanes with contour stellar based method is only 16.53%.

From Fig. 22, we can see that the recognition accuracy of 21 airplanes with complex convolutional neural network based method is only 67.53%.

From Fig. 23, we can see that the recognition accuracy of 21 airplanes with deep residual network based method is only 4.34%.

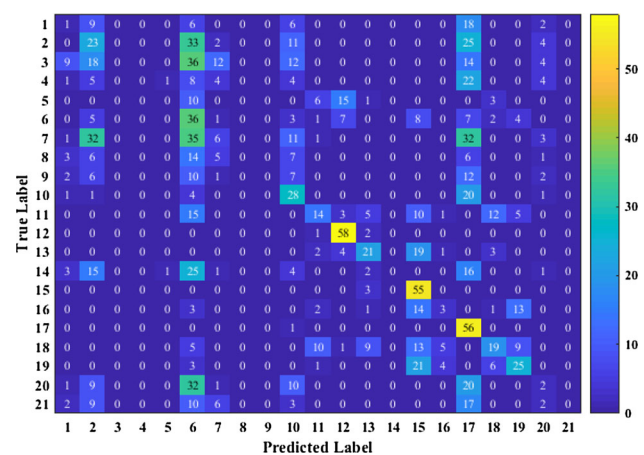


Fig. 21 The recognition accuracy of 21 airplanes on the basis of contour stellar based method

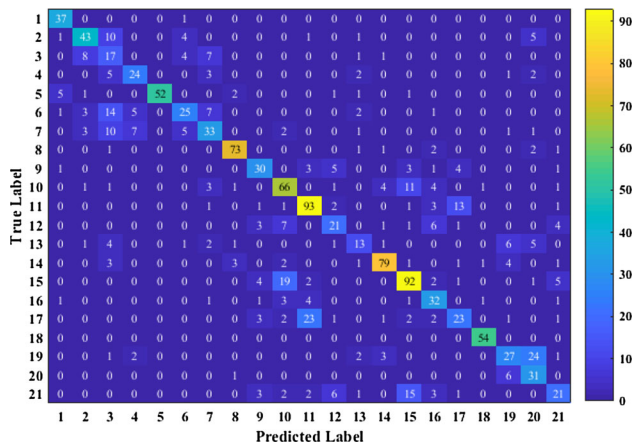


Fig. 22 The recognition accuracy of 21 airplanes on the basis of complex convolutional neural network based method

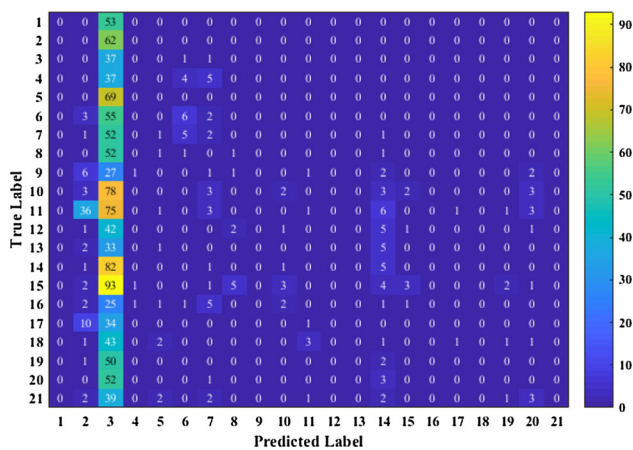


Fig. 23 The recognition accuracy of 21 airplanes on the basis of deep residual network based method

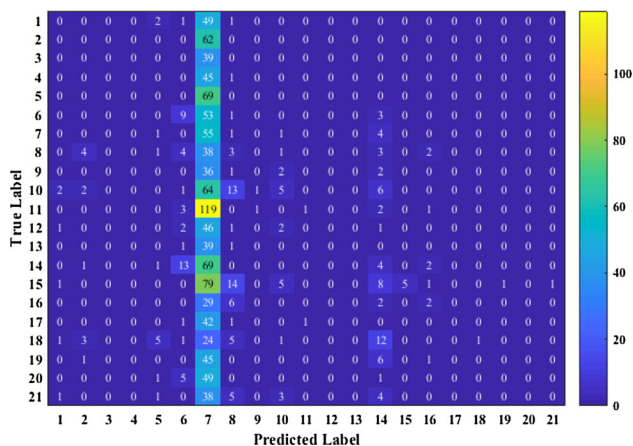


Fig. 24 The recognition accuracy of 21 airplanes on the basis of deep complex residual network based method

From Fig. 24, we can see that the recognition accuracy of 21 airplanes with deep complex residual network based method is only 6.48%.

From Table 3, and above figures, we can also see that, the proposed method with slice combination and feature fusion shows the great improvement of the recognition accuracy compared with the existing fingerprint recognition technology based on traditional machine learning and the existing fingerprint recognition technology based on deep learning. And when the number of airplanes increases, the recognition accuracy of the proposed method with slice combination and feature has barely dropped, remaining above 99%.

Further, by artificially adding Gaussian white noise to the measured dataset, the recognition accuracy of the method proposed in this paper can be further tested under different signal-to-noise ratios, as shown in Fig. 25.

From Fig. 25, we can see that even if the signal-to-noise ratio drops to -5 dB, the recognition accuracy for a total of 198 airplanes can still reach more than 91%. And therefore the designed slice combination convolutional neural network model has greatest application potential for large-scale real-world RFF Identification for Internet of Things.

5 Conclusion and discussion

It is an important issue for researchers to establish a suitable deep learning model for radio frequency fingerprint identification under deep learning based communication framework. In order to prevent the occurrence of problems such as device cloning, replay attacks and user identity impersonation, given the limitations of existing methods, this paper proposed a novel slice combination convolutional neural network model for large-scale real-world RFF recognition. Through detailed experimental tests, the following conclusions are obtained:

- (1) The proposed method can effectively guide the deep neural network structure to learn the real RF fingerprint characteristics, by signal slice combination, feature combination and fusion.
- (2) The proposed method shows great improvement of the recognition accuracy with both the training accuracy and prediction accuracy over 99%, compared with existing fingerprint recognition techniques.
- (3) The proposed method has great application potential for large-scale real-world individual identification of

Table 3 Comparative recognition results by these methods

Method	No. of airplanes	Training accuracy	Prediction accuracy (%)
PSD & GRA	21	–	82.09
PSD & SVM	21	–	91.23
Contour stellar [22]	21	35.96%	16.53
Residual network [19]	21	22.60%	4.34
Complex-value residual network [19]	21	45.71%	6.48
Complex-value convolutional neural network [30]	21	80.79	67.53
Proposed method with single slice	21	95.20%	92.68
Proposed method with two slices	21	99.89%	99.70
PSD & SVM	198	–	68.31
Proposed method with single slice	198	79.88%	74.60
Proposed method with two slices	198	99.55%	99.03

The proposed method with only single slice is the deep network without slice combination and feature fusion; The proposed method with two slices is the deep network with combination of two slices and feature fusion under the same deep convolutional neural network feature extractor structure. Each signal slice contains 1500 steady-state data points

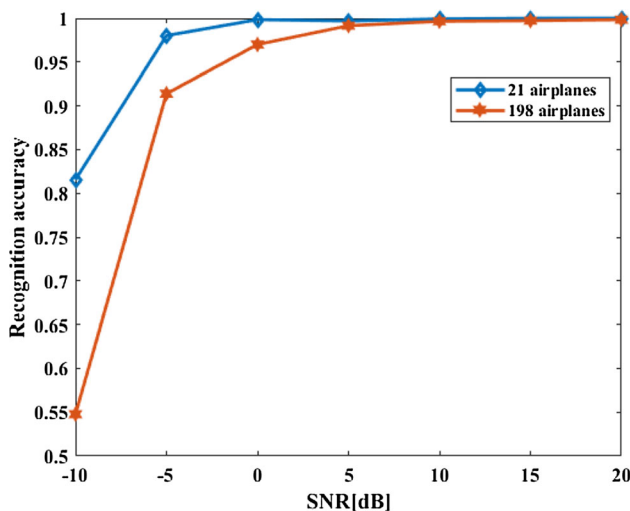


Fig. 25 The recognition accuracy under different signal-to-noise ratios by the proposed method

communication radiation sources for Internet of Things.

In the future research work, we will continue to optimize the model structure and model parameters of the proposed method.

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References

- Wang, M., Lin, Y., Tian, Q., & Si, G. (2021). Transfer learning promotes 6g wireless communications: recent advances and future challenges. *IEEE Transactions on Reliability*, 70(2), 1–18. <https://doi.org/10.1109/TR.2021.3062045>
- Bao, Z., Lin, Y., Zhang, S., Li, Z., & Mao, S. (2021). Threat of adversarial attacks on DL-based IoT device identification. *IEEE Internet of Things Journal*, 9(11), 9012–9024. <https://doi.org/10.1109/JIOT.2021.3120197>
- Hou, C., Liu, G., Tian, Q., Zhou, Z., Hua, L., & Lin, Y. (2022). Multi-signal modulation classification using sliding window detection and complex convolutional network in frequency domain. *IEEE Internet of Things Journal*. <https://doi.org/10.1109/JIOT.2022.3167107>
- Dong, Y., Jiang, X., Zhou, H., Lin, Y., & Shi, Q. (2021). SR2CNN: Zero-shot learning for signal recognition. *IEEE Transactions on Signal Processing*, 69, 2316–2329. <https://doi.org/10.1109/TSP.2021.3070186>
- Liu, S., Wang, S., Liu, X., et al. (2021). Human memory update strategy: A multi-layer template update mechanism for remote visual monitoring. *IEEE Transactions on Multimedia*, 23, 2188–2198. <https://doi.org/10.1109/TMM.2021.3065580>
- Wang, S., Liu, X., Liu, S., et al. (2022). Human short-long term cognitive memory mechanism for visual monitoring in IoT-assisted smart cities. *IEEE Internet of Things Journal*, 9(10), 7128–7139. <https://doi.org/10.1109/JIOT.2021.3077600>
- Li, J., Ying, Y., & Ji, C. (2019). Study on radio frequency signal gene characteristics from the perspective of fractal theory. *IEEE Access*, 7, 124268–124282. <https://doi.org/10.1109/ACCESS.2019.2938791>
- Lin, Y., Zhao, H., Ma, X., Ya, Tu., & Wang, M. (2021). Adversarial attacks in modulation recognition with convolutional neural networks. *IEEE Transactions on Reliability*, 70(1), 389–401. <https://doi.org/10.1109/TR.2020.3032744>
- Chen, X., Li, J., Han, H., & Ying, Y. (2018). Improving the signal subtle feature extraction performance based on dual improved fractal box dimension eigenvectors. *Royal Society Open Science*, 5(5), 1–10. <https://doi.org/10.1098/rsos.180087>
- Zheng, T., Sun, Z., Ren, K. (2019). FID: Function modeling-based data-independent and channel-robust physical-layer identification [J/OL]. [2019-01-17]. arXiv preprint [arXiv:1901.05914v1](https://arxiv.org/abs/1901.05914v1), 2019. <https://arxiv.org/abs/1901.05914v1>
- Li, J., Li, Y., Shouhei, K., & Tetsuo, K. (2013). A robust signal recognition method for communication system under time-

- varying SNR environment. *IEICE Transactions on Information and Systems*. <https://doi.org/10.1587/transinf.E96.D.2814>
12. Shu, Y., Huang, Y., Zhang, J., et al. (2016). Gradient-based fingerprinting for indoor localization and tracking. *IEEE Transactions on Industrial Electronics*, 63(4), 2424–2433. <https://doi.org/10.1109/TIE.2015.2509917>
 13. Lin, K., Chen, M., Deng, J., et al. (2016). Enhanced fingerprinting and trajectory prediction for IoT localization in smart buildings. *IEEE Transactions on Automation Science and Engineering*, 13(3), 1294–1307. <https://doi.org/10.1109/TASE.2016.2543242>
 14. Li, J., Bi, D., Ying, Y., et al. (2019). An improved algorithm for extracting subtle features of radiation source individual signals. *Electronics*, 8(2), 1–11. <https://doi.org/10.3390/electronics8020246>
 15. Han, H., Li, J., & Chen, X. (2018). The individual identification method of wireless device based on a robust dimensionality reduction model of hybrid feature information. *Mobile Networks & Applications*, 23(4), 709–716. <https://doi.org/10.1007/s11036-018-1003-5>
 16. Li, J. (2015). A novel recognition algorithm based on holder coefficient theory and interval gray relation classifier. *KSII Transactions on Internet and Information Systems (TIIS)*, 9(11), 4573–4584. <https://doi.org/10.3837/tiis.2015.11.018>
 17. Li, J. (2015). A new robust signal recognition approach based on holder cloud features under varying SNR environment. *KSII Transactions on Internet and Information Systems (TIIS)*, 9(12), 4934–4949. <https://doi.org/10.3837/tiis.2015.12.011>
 18. Yuan, Y., Huang, Z., Hao, Wu., et al. (2014). Specific emitter identification based on Hilbert-Huang transform-based time-frequency-energy distribution features. *IET communications*, 8(13), 2404–2412. <https://doi.org/10.1049/iet-com.2013.0865>
 19. Wang, S., Jiang, H., Fang, X., Ying, Y., Li, J., & Zhang, B. (2020). Radio frequency fingerprint identification based on deep complex residual network. *IEEE Access*, 8, 204417–204424. <https://doi.org/10.1109/ACCESS.2020.3037206>
 20. Jiang, Yu., Peng, L., Aiqun, Hu., et al. (2019). Physical layer identification of LoRa devices using constellation trace figure. *EURASIP Journal on Wireless Communications and Networking*, 1, 1–11. <https://doi.org/10.1186/s13638-019-1542-x>
 21. Peng, L., Zhang, J., Liu, M., & Hu, A. (2020). Deep learning based RF fingerprint identification using differential constellation trace figure. *IEEE Transactions on Vehicular Technology*, 69(1), 1091–1095. <https://doi.org/10.1109/TVT.2019.2950670>
 22. Ying, Y., Li, J., Ji, C., et al. (2021). Differential contour stellar based radio frequency fingerprint identification for internet of things. *IEEE Access*, 9, 53745–53753. <https://doi.org/10.1109/ACCESS.2021.3071352>
 23. Ding, L. D., Wang, S. L., Wang, F. G., & Zhang, W. (2018). Specific emitter identification via convolutional neural networks. *IEEE Communications Letters*, 22(12), 2591–2594. <https://doi.org/10.1109/LCOMM.2018.2871465>
 24. Tu, Y., Lin, Y., Wang, J., & Kim, J. U. (2018). Semi-supervised learning with Generative Adversarial Networks on digital modulation classification. *CMC-Computers, Materials & Continua*, 55(2), 243–254. <https://doi.org/10.3970/cmc.2018.01755>
 25. Merchant, K., Revay, S., Stantchev, G., et al. (2018). Deep learning for RF device fingerprinting in cognitive communication networks. *IEEE Journal of Selected Topics in Signal Processing*, 12(1), 160–167.
 26. Li, J., Ying, Y., & Ji, C. (2019). Study on gas turbine gas-path fault diagnosis method based on quadratic entropy feature extraction. *IEEE Access*, 7, 89118–89127. <https://doi.org/10.1109/ACCESS.2019.2927306>
 27. Lukacs, M. W., Zeqolari, A. J., Collins, P. J., et al. (2015). “RF-DNA” fingerprinting for antenna classification. *IEEE Antennas and Wireless Propagation Letters*, 14, 1455–1458. <https://doi.org/10.1109/LAWP.2015.2411608>
 28. Polak, A. C., & Goeckel, D. L. (2015). Wireless device identification based on RF oscillator imperfections. *IEEE Transactions on Information Forensics and Security*, 10(12), 2492–2501. <https://doi.org/10.1109/TIFS.2015.2464778>
 29. Ya, Tu., Lin, Y., et al. (2022). Large-scale real-world radio signal recognition with deep learning. *Chinese Journal of Aeronautics*, 35(9), 35–48. <https://doi.org/10.1016/j.cja.2021.08.016>
 30. Ying, L., Li, J., & Zhang, B. (2021). Differential complex-valued convolutional neural network-based individual recognition of communication radiation sources. *IEEE Access*, 9, 132533–132540. <https://doi.org/10.1109/ACCESS.2021.3114191>

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